MIL-H-51182A(EA)
11 March 1988
SUPERSEDING
MIL-H-51182(MU)
28 February 1964

MILITARY SPECIFICATION

HEADPIECE, HOSPITAL COLLECTIVE PROTECTOR, M13A1

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a vinyl plastic headpiece.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 101 - Test Procedures for Packaging Materials.
FED-STD-123 - Marking for Shipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSMC N/A FSC 4240

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-70-23 - Headpiece, Hospital Collective Protector, M13A1.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-70-23 - Headpiece, Hospital Collective Protector, M13A1.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contacting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicatation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM STANDARDS

D 882 - Tensile Properties of Thin Plastic Sheeting.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:52Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-70-23 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the headpiece shall conform to the specifications and drawings listed on Drawing 5-70-23 and subsidiary drawings.

3.2 Manufacture and assembly. The headpiece shall be manufactured and assembled as specified on Drawing 5-70-23.

3.3 Seam strength.

3.3.1 Hood seams. The seam shall withstand a minimum pull of 12 pounds per inch when tested as specified in 4.3.1 at a steady pull rate of 2 inches per minute.

3.3.2 Loop attachments. The loop shall withstand a minimum pull of 20 pounds without tearing or separation when tested as specified in 4.3.2 at a steady pull rate of 2 inches per minute.

3.4 Workmanship. The headpiece shall have neat, uniform and continuous seams and shall be free from wrinkles, punctures and tears and foreign matter such as dirt, oily and viscous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the packaged hoods offered for acceptance at one time which have been produced by one manufacturer at one plant under essentially the same manufacturing conditions and techniques with no change in materials.

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.3 For destructive test. Sampling of the headpiece shall be in accordance with level S-3 of MIL-STD-105. Sampling of packaging shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample headpieces and packaging shall be examined and tested in accordance with the classification of defects (4.2.3.3) and MIL-STD-105.

4.2.3.2 For nondestructive test. Each packaged headpiece shall be examined in accordance with 4.2.3.3.

4.2.3.2.1 For destructive test.

(a) Packaging. Packaged headpieces shall be tested in accordance with 4.3.4 and MIL-STD-105, using an AQL of 2.5 percent defective.

(b) Headpieces. Sample headpieces shall be tested in accordance with 4.3.1, 4.3.2, and MIL-STD-105, using and AQL of 4.0 percent defective.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:52Z
Check the source to verify that this is the current version before use.

4.2.3.3 Classification of defects.

(a) Headpiece, hospital collective protector, M13A1 (Drawing 5-70-23).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing or incorrectly assembled
102	Tape not applied as specified
103	Thread incorrect
104	Marking incorrect
105	Workmanship (see 3.4)
Minor:	AQL 2.5 percent defective
201	Marking illegible

(b) Packaging inspection (SPI P5-70-23).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Item not completely cleaned/dried	
102	Unit and supplemental container and closure damaged or not as specified	
103	Unit or supplemental container marking missing, incorrect or illegible	
104	Unit pack container leakage	4.3.3

4.3 Tests. Tests shall be conducted as follows:

4.3.1 Hood seam strength. The bond strength of the seam shall be determined in accordance with ASTM D 882. In determination of the bonding strength, the seam shall be at the center of the specimen and perpendicular to the center line. At least five specimens shall be tested for bond strength. The average of the bond tensile strength shall be specified.

4.3.2 Loop attachment strength. The loop attachment shall be tested for strength using one 2- by 6-inch specimen for each loop from one-half of the sample headpieces. A testing machine of the constant rate-of-jaw-separation type shall be used for this test. The loop shall be at one end of the specimen and the 6-inch dimension shall be parallel with the long dimension of the loop. One of the grips of the testing apparatus shall be replaced by a rotating mandrel,

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:52Z
Check the source to verify that this is the currentversion before use.

approximately 1/2 inch in diameter, from which the loop shall be suspended. Condition the specimen (or headpiece prior to cutting) for a minimum of one-half hour at 70° ± 5°F immediately prior to testing.

4.3.3 Unit pack container leakage (SPI P5-70-23). The leakage of the unit container (see MIL-B-117 container only), containing the item shall be determined by subjecting the samples to the hot water technique of method 5009.2 of FED. TEST METHOD STD. NO. 101.

4.3.4 Heat seal seam separation (SPI P5-70-23). The heat seal seams of the unit pack container shall be tested in accordance with FED. TEST METHOD STD. NO. 101, method 2024, except that the static load shall be 45 +5, -0 ounces, and a 5 percent reduction shall be permitted when room temperature in test area exceeds 90°F.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI P5-70-23.

5.2 Interplant shipment (see 6.3). The headpiece shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in accordance with FED-STD-123, and shipping containers shall include the contractors name, address, contract number, and date of manufacture.

5.3 Repair parts. When this item is being procured for storage and issue as a repair part, unit packing, packing and marking shall be as specified in SPI P5-70-23.

6. NOTES

6.1 Intended use. The M13A1 headpiece is intended to provide protection for casualties who are unable to wear a protective mask.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Unit packing and packing required (see section 5).
(c) Shelf life code.

6.3 Interplant shipment. Packaging and packing for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or prime contractor or from a vendor to contractor to a military arsenal or plant.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:52Z
Check the source to verify that this is the current version before use.

6.4 Changes from previous issue. Asterisks and vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term (key word) listing.

Headpiece, plastic
Hospital, collective protection equipment
Protector, chemical-biological-radiological agents

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A939

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-51182A(EA)	HEADPIECE, HOSPITAL COLLECTIVE PROTECTOR, M13A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:52Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in *submitting comments and suggestions for improvements.* All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 Alexandria, VA

POSTAGE WILL BE PAID BY

Commander
U.S. Army Chemical Research, Development
and Engineering Center
ATTN: SMCCR-SPT-S
Aberdeen Proving Ground, MD 21010-5423

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.